TECHNICAL APPENDIX: R/PLACE AS A MODEL ORGANISM

Model organisms let separate studies build one account of the same organism (Ankeny & Leonelli, 2011). Research on the fruit fly relates genetics, development and physiology within the whole organism (Adams et al., 2000). r/place has so far been studied in parts. Rappaz et al. (2018) examined collaboration among users, Israeli et al. (2022) community participation, and Jemine (2025) coordination in four communities. Each answers its own question, and nothing yet relates their findings.

As a model organism, r/place turns these studies into parts of one agenda, which asks how communities recruit members, coordinate work, maintain artworks and deal with their neighbors. Our instrument composes coded accounts of 705 curated communities into a population model in which these processes appear as parts of the same organizing (Section 2). Later studies can place their findings against this baseline.

Table 1 compares r/place with the fruit fly and with two research objects outside biology, chess and ImageNet 2012. Its rows adapt four features that Ankeny and Leonelli (2011) discuss in model organism research. Material accessibility means that researchers can obtain and examine the same material. Integrative understanding means that studies of different processes add to one account of the whole organism. Standardization means that shared conditions make specimens or cases comparable. A research community shares materials, methods and results, so later work can revisit and revise earlier findings. Chess (Simon & Chase, 1973) and ImageNet (Deng et al., 2009) show that research objects built by people can carry some of these features, though ImageNet supports common evaluation more than integration across phenomena. Each column shows how one research object meets the four features. Sections 1 to 4 take them in turn for r/place; Section 5 describes the instrument.

1. MATERIAL ACCESSIBILITY

The 72-hour event provides a bounded record of communities forming, building and defending artworks. Reddit discussions can be read alongside pixel changes (Section 5.1). Researchers can return to the same evidence instead of relying only on published accounts. LLM agents reduce the cost of reading the 705

communities' records (Section 5.2). The dataset is frozen, but researchers can change the instrument, use another literature or broaden the census within the dataset.

2. INTEGRATIVE UNDERSTANDING

The aim is to relate processes that separate studies can examine on their own. From our preliminary findings, we observe that recruitment, coordination, maintenance and relations with neighbors form distinctive roles in how communities organize (Table 2; Section 5.3). The current model provides a basis for comparison with open-source communities and self-managing firms (O'Mahony & Ferraro, 2007; Lee & Edmondson, 2017). Few communities show a clear signal of authority, such as a leader, a vote, an oath, a sanction or a division of work. The coders found one in 44 of the 705 communities while coding, and in about 119 when the returned model asked the community coders to check explicitly for it. These indicators describe different arrangements and do not form a single measure of formal authority. Extending the model to hierarchical firms requires further evidence. Later researchers can examine other connections or challenge those proposed.

Table 1. Features of model organism research and related practices in chess, ImageNet 2012 and r/place.

Characteristic	Fruit fly	Chess	ImageNet 2012	r/place
Field	Genetics	Cognitive psychology	Machine learning	Organization science
Material accessibility	Standard fly strains that laboratories obtain, breed and examine.	A game with explicit rules, recorded positions and recorded games.	Labeled images in 1,000 categories, available for research download.	Public records of pixel changes and Reddit discussions.
Integrative understanding	Relates genetics, development and physiology within the whole organism.	Relates perception, memory and search in skilled play.	Compares classification and localization methods on a shared image collection.	Relates recruitment, coordination, maintenance and relations among communities.
Standardization	Standard strains and shared protocols make specimens comparable.	Fixed rules and measurable differences in skill.	Agreed labels, data splits and evaluation rules.	Shared event rules; coding contracts held constant within a run.
Research community	Stock centers and FlyBase support sharing specimens, methods and results.	Published position tasks and game records let later work revisit findings.	Development kits and a public challenge put new methods to common tests.	Shared coded cases, contracts and checkpoints invite reuse and revision.

3. STANDARDIZATION

All communities worked under the same pixel-placement rules: each user could change one pixel's color every five minutes, and anyone could overwrite it (Simpson et al., 2017). Communities nevertheless differed in their histories and circumstances. We held the coders' contract constant within a run (Section 5.4). Later runs can change a specified component, such as the literature, community coder or aggregator, and compare the results with the baseline. The 298 communities coded on two LLMs give a first measure of variation between models. A planned run with an open-weight model will extend that comparison and lower the cost of replication.

4. RESEARCH COMMUNITY

A model organism becomes a common research resource through sustained sharing and reuse (Ankeny & Leonelli, 2011; Kohler, 1994). We provide the baseline, coded cases and tools for inspecting the analysis (Section 5.5). Stable identifiers link claims back to their evidence, so later researchers can compare, revise or replace accounts and share the changes. This provides infrastructure for a research community; whether one forms depends on others adopting and extending it. Researchers can rerun the instrument or resume it from a checkpoint, then return a revised population model to the community coders to assess its fit to their cases.

5. INSTRUMENT

The instrument addresses the three constraints drawn from Machlup (1967): public records make behavior observable, LLMs reduce the cost of reading, and the aggregator composes the coded cases.

5.1 Data and census

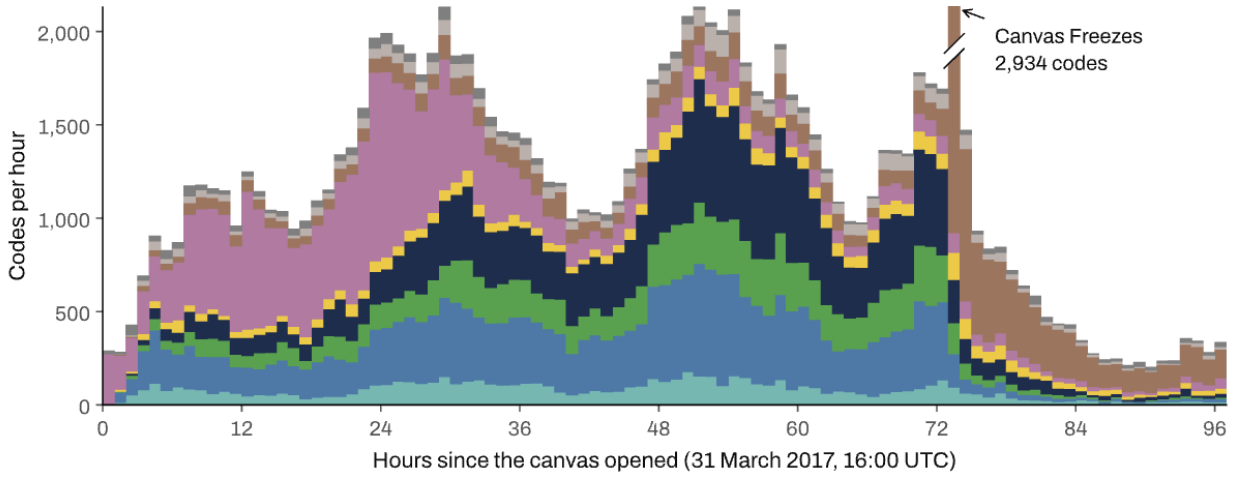
The dataset joins four sources. Reddit's log records the time, position, color and hashed user ID of every pixel change: 16,559,897 changes by 1,166,924 users (Reddit, 2017). The Arctic Shift archive holds Reddit's posts and comments (Heitmann, n.d.). For each subreddit, we collected the posts and comments from the 72 hours of the event and the 24 hours after. A table on the Internet Archive links the hashed user IDs to usernames, allowing pixel changes and posts to be joined. The Place Atlas identifies the subreddits associated with each artwork (Place Atlas, 2017).

The wider dataset contains 7.5 million posts and comments across 2,796 subreddits. Our census is drawn from the Place Atlas (2017), which credits 777 subreddits with 1,236 of the 1,473 artworks. We retained subreddits whose members wrote about r/place while the canvas was open, our criterion for evidence of coordination. This left 706 subreddits, of which we excluded one[1]. The 705 communities in the census wrote 2,130,302 posts and comments and account for 90% of all pixel changes. The repository records the reason for each exclusion.

5.2 Coding and the population model

The harness assigned one community coder to each of the 705 communities. Each coder read its community's record in time order, from the opening of the canvas to a day after it froze, and coded what members did in the members' own words. Coders with large records split them into sections and passed these to further coding agents. The coders produced 115,388 first-order codes and grouped them into 7,844 second-order themes (Gioia et al., 2013). Figure 1 counts the codes by the hour of the post or comment each one cites, colored by the aggregate dimension it later falls under. Each coder then wrote a case account of its community for cross-case analysis (Eisenhardt, 1989).

Figure 1. First-order codes per hour across the 705 communities, by aggregate dimension. Hours count from 31 March 2017, 16:00 UTC. The bar after the freeze (2,934 codes) is cut.



[1] We left out one community, r/The_Donald, for two reasons. First, Reddit banned it in June 2020 for repeatedly hosting content that broke its rules, the first of which bans promoting hate based on identity or vulnerability (Reddit, 2020). Second, its exclusion has little bearing on the model: its single artwork covered 108 pixels, within the lowest quartile of the census, and was largely overwritten before the canvas froze (Place Atlas, 2017).

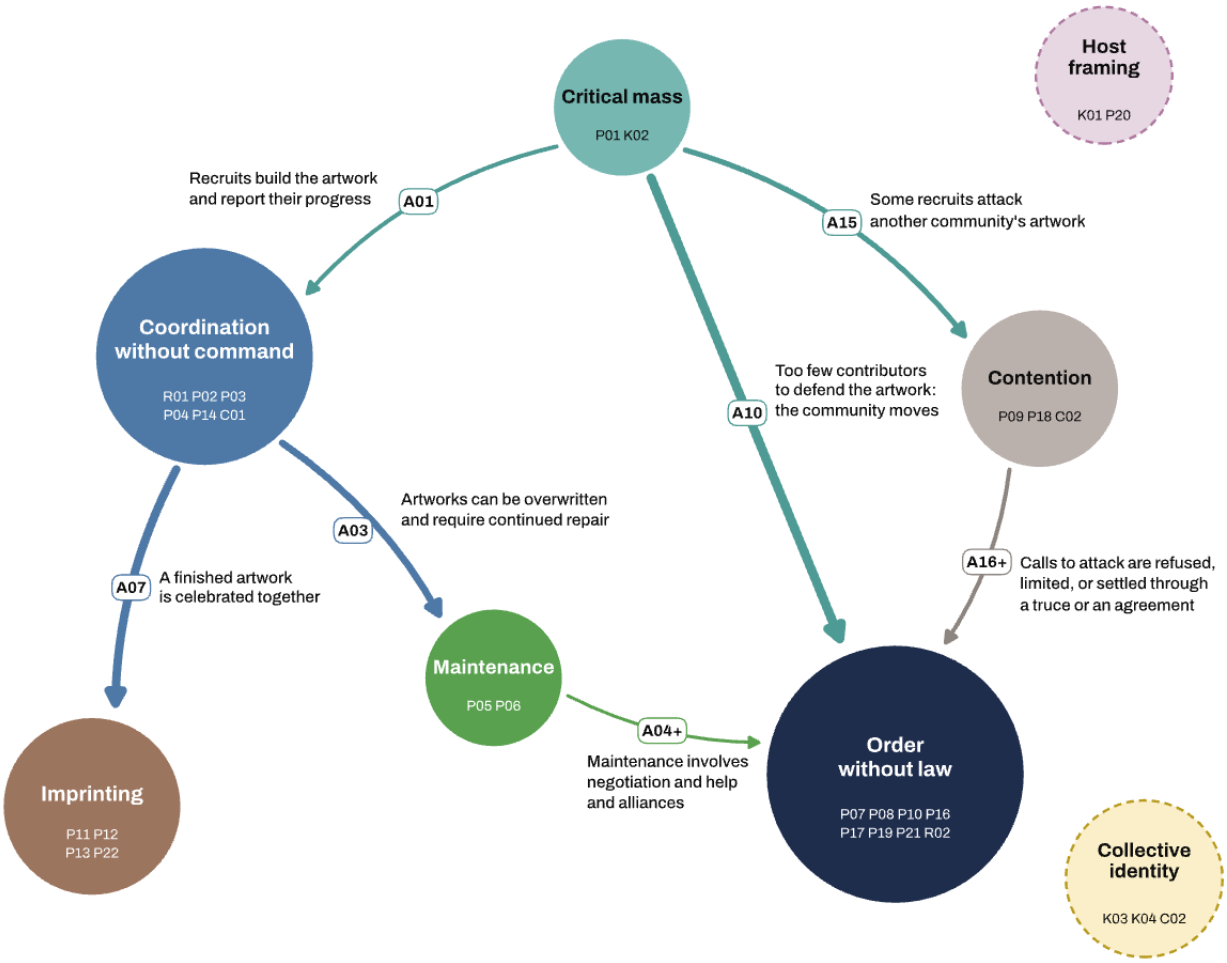
Table 2. The eight dimensions, their content in the model and related literature. Dimension colors match the figures. Numbers are first-order codes in each dimension within the 97 hours of Figure 1; 2,476 codes have no dimension.

Dimension	What it covers in the model	Related literature
Coordination without command (23,408 instances)	How communities coordinate work on a shared artwork through design choices, progress reports, templates, scripts and chat, with occasional use of a leader or a vote.	Faraj & Xiao (2006)
Maintenance (13,049 instances)	How communities maintain completed artworks through repair and distinguish attacks from changes by allies or their own mistakes.	Graham & Thrift (2007)
Order without law (22,433 instances)	How communities manage claims to space and exchange help without authority over one another, including restraint, negotiated settlements and alliances.	Ellickson (1991); Ring & Van de Ven (1994)
Collective identity (4,892 instances)	How communities express collective identities through their artworks, where those identities come from, and what they defend or reject on the canvas	Melucci (1995)
Imprinting (13,674 instances)	Legacies recorded during and immediately after the event: celebrations, canvas records, new emblems, intercommunity ties and new members. Persistence beyond the observation window remains to be tested.	Marquis & Tilcsik (2013)
Host framing (23,589 instances)	How existing communities frame and accommodate the event within their subreddit, including treating it as play, limiting its presence, or both.	Fiss & Zajac (2006)
Critical mass (7,289 instances)	How communities recruit participants, teach them to place pixels, and judge whether they have enough contributors to build their artwork before the event ends.	Marwell & Oliver (1993)
Contention (4,578 instances)	How communities challenge another community's artwork and sustain the contest through actions against the work and discussion that identifies the other community as an opponent.	Tilly & Tarrow (2007)

The aggregator grouped the second-order themes into 197 third-order themes, 28 fourth-order themes and eight aggregate dimensions (Table 2). We retain its labels to keep the text, figures and archived model aligned. The references identify related literature; the table states what each dimension covers in this model. The aggregator held this coding fixed while building and revising the population model. The model links elements such as processes, conditions and roles. Figure 2 draws the model across the eight dimensions (Gioia et al., 2013), keeping the arrows whose time order holds across communities. A call for hands starts the sequence (critical mass). Recruits build the artwork and report their progress (coordination without command). Because others overwrite it, building turns into repair (maintenance), and a finished artwork is celebrated together (imprinting). Three arrows end in order without law. Keeping an artwork

leads to negotiation, help and alliances with neighbors. Calls to attack a rival are refused, limited or settled by truce. A community with too few hands moves. Host framing and collective identity hold elements but no arrow: their elements are mostly conditions that frame the event rather than steps within it.

Figure 2. The population model after the return, across eight aggregate dimensions. Circle size reflects the number of model elements listed inside it. An arrow links the dimensions of two connected elements and is drawn where the source dimension comes first in more communities than in 95% of 500 label swaps. Dashed circles have no such arrow. A04+ combines A04, A13 and A19; A16+ combines A16 and A17.



5.3 Model revision and organizational forms

The aggregator returned the population model to every community coder. Each coder checked it against its community's record and proposed changes to individual elements: instantiating or marking an element absent, adding conditions, steps or links, restricting or contradicting a claim, or adding an element (Fisher & Aguinis, 2017). The coders proposed over 42,000 changes. The aggregator reviewed them and revised the model while keeping the coding fixed. Before revision, the model fit 201 communities fully and 489 in

part; afterward, it fit 630 fully and 56 in part. The same agents produced the coding and assessed fit, so these results are an internal check. Section 5.4 separates completed checks from planned assessments.

The requested changes also distinguish recurring configurations of organizing. We pair hierarchical clustering with profiles of the resulting groups, as in Awad et al.'s (2018) Moral Machine study. Figure 3 groups communities by their selected changes and their codes; Figure 4 compares the six clusters across the eight dimensions. We treat these clusters as candidate organizational forms, whose interpretation requires examining how communities solve organizing problems (Puranam et al., 2014). In form C, the design reopened, 40.3% of changes concern coordination without command, compared with 27.6% across communities. Each form retains its community identities, so its interpretation can be checked against their codes and quotations.

Figure 3. Six candidate organizational forms. Ward clustering groups the 558 communities with five or more selected changes, with changes weighted two thirds and codes one third. The aggregator chose the cut into six groups and placed the remaining 147 communities beside a community it judged similar. The dashed circle marks the cut. Bars outside the tree show each community's selected changes.

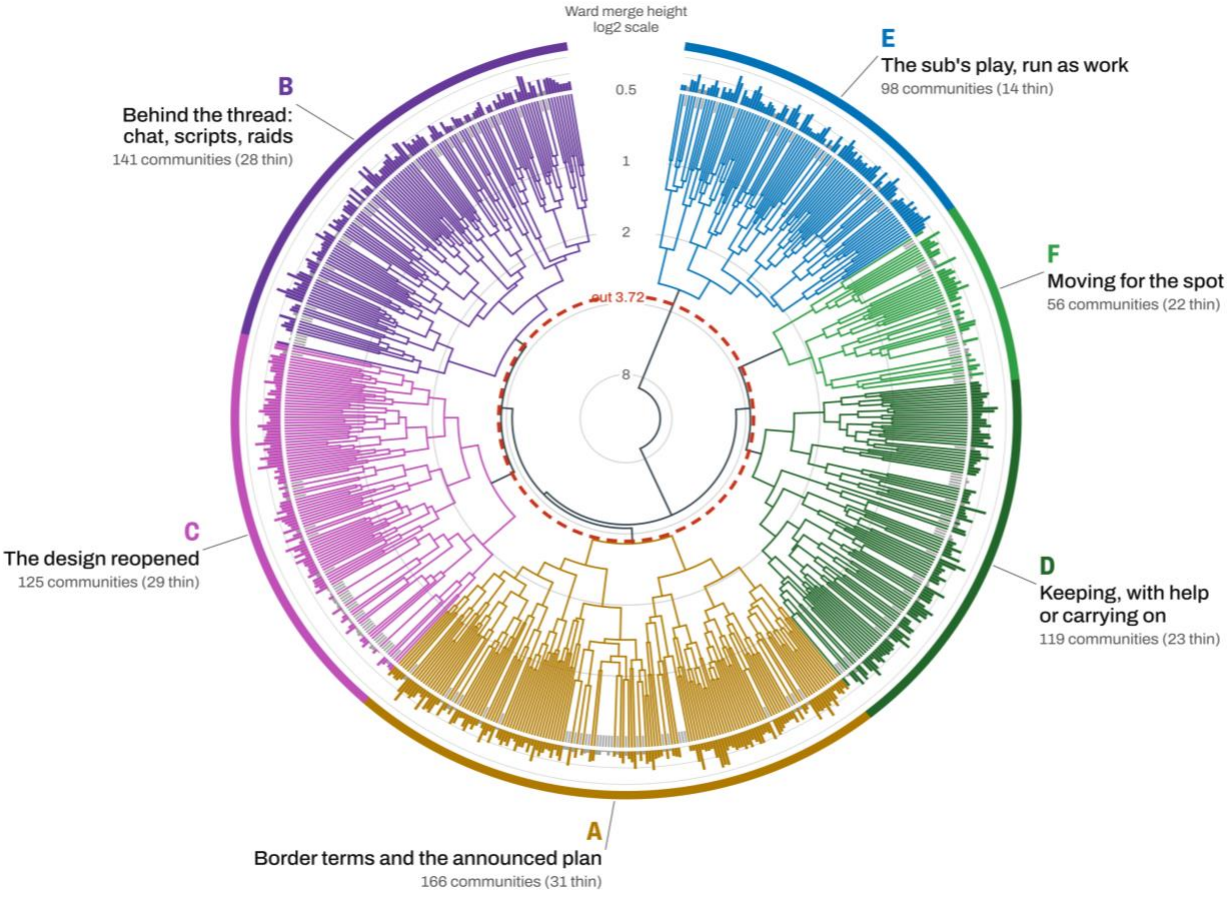


Figure 4. Requested model changes by candidate organizational form. Each line shows a form's mean share of selected changes in each dimension, divided by the mean among the 684 communities with changes. Bands show 95% confidence intervals. The dashed circle marks the average community. Rings double from ×0.25 at the center to ×4 at the rim.



Similar requests may reflect omissions in the population model as well as differences in organizing. The case accounts must therefore show shared ways of organizing work before the groups can be treated as forms. Weighting changes more heavily makes departures from the model the main basis of grouping. The cut into six groups and the placement of 147 communities outside the clustering are analytical choices. We will assess sensitivity to the cut and weights, and examine the assigned communities separately.

5.4 Contracts and checks

Written contracts specify what each agent reads, produces and may do, and how its work is checked. A community coder reads only its own community's record through a tool that logs the material provided; it may not open the atlas or earlier coding. The aggregator reads the community coders' sealed work, never

the source record. The aggregator composes descriptions; we judge whether the resulting model makes a theoretical contribution.

We developed the contracts over many runs on the same material. Earlier versions coded 347 communities in 462 runs. The final coding of the 705 communities used v32 of the contract. The aggregator has run four times, each with more data and updated models. Between runs we interviewed the agents. When their work departed from the grounded-theory method (Gioia et al., 2013), we revised the contract. We retained a version when its output was internally coherent and the agent could explain its assigned task.

The contracts include three safeguards. Agents cite source references rather than typing quotation text. They also distinguish absence from missing evidence: when an element does not appear in a community's record, the coder classifies it as absent, not needed, unobserved or not testable. Only the first two count against the model. An agent may depart from its contract only after documenting the departure, and our own rulings are recorded alongside its work.

Completed checks include automated source checks for code quotations, review of the aggregator's work by a separate agent, and interviews during contract development. Reliability will be assessed by comparing the paired coding of 298 communities. Further checks include human audits of selected inferences, two alternative population models, comparison with the excluded r/place subreddit record, and an open-weight rerun. The r/place agent's preliminary reading already added two elements and an arrow to the model. Completing that reading will provide further triangulation; we will distinguish evidence already used in revision from evidence examined afterward.

5.5 Sharing

The harness saved session snapshots throughout the analysis[2]. A checkpoint interview restores a saved agent session and its matching files (https://os-fruit-fly.github.io/rplace/interviews.html). Readers can question its decisions and compare new answers with the recorded work. The repository provides 18 aggregator

[2] The checkpoints were stored in a non-anonymized repository. There was a total of 320 checkpoints committed to the repository during the 18 stages of the aggregator.

checkpoints, one at each stage close, and 24 community checkpoints from 12 communities at their coding close and end of building their local model.

Readers can open a checkpoint interview in two ways. “Chat Locally” gives a terminal command that downloads the checkpoint and opens the interview in the reader’s own Claude Code or Codex CLI. “Chat Online” lets reviewers interview agents in the browser, with no subscription or download, after signing in at https://os-fruit-fly.github.io/rplace/open-interviews.html.

The repository also holds download scripts, contracts and every community’s coded record. The full research repository, with the record of every upload, can be shared with the editor on request. Usernames are used only to join the sources and are omitted from shared outputs. We provide scripts to retrieve the source records rather than redistribute the corpus. Ethics review is underway.

Researchers can rerun the instrument or resume it from a checkpoint. A rerun can add a mechanism the current model lacks by handing the aggregator a new literature. For example, r/PlaceNL coordinated work for r/TheNetherlands, its parent subreddit. However, the current model does not consider literature on coordination across communities. A run that keeps the current coding scheme and adds a literature on interorganizational coordination needs to rerun only 3 of the aggregator’s 18 stages to produce a population model. In this run, those three stages took under six hours of the aggregator’s working time (725 million tokens, about US$220 at list prices), and returning the new model to all 705 coders took about ten hours more.

6. LIMITATIONS

r/place limits which organizational claims can be studied in it. The event offers no setting for studying pay, employment contracts, careers, long organizational horizons or costly exit. The current model gives limited evidence about formal authority. Claims about these features require other settings and records, such as GitLab’s public record (Choudhury et al., 2020) or Enron’s publicly available internal mail (Klimt & Yang, 2004).

The record also has gaps. It covers Reddit discussions and pixel changes, but not the Discord servers where some communities also met. The census starts from the curated atlas of the finished canvas. A

community whose artwork was wholly erased enters only if a surviving artwork also credits it, so failure is undersampled (Denrell, 2003).

Further work should test how the population model depends on the literature, event and instrument. Reading the record against another literature can show which findings depend on the concepts used. The 2022 and 2023 events can test the model's scope where the relevant rules are comparable. Rerunning the 2017 analysis with another instrument, including the planned open-weight model, can test which findings can be recovered. Differences must be traced to the evidence, scope or analytical choices before deciding which findings to revise or qualify.

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